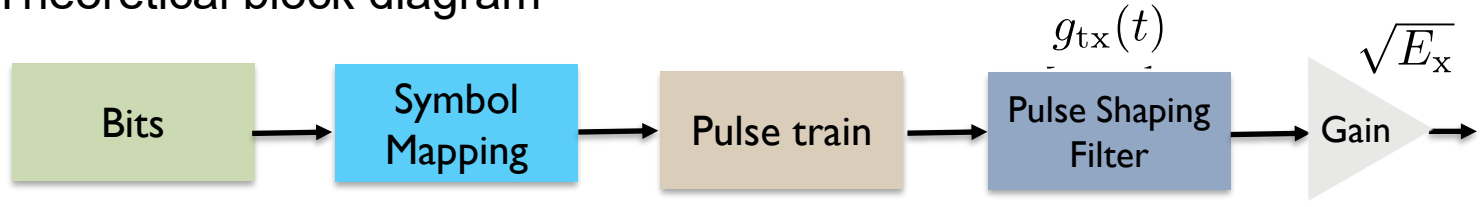
Lecture #9

EE 471C / EE 381K-17 Wireless Communication Lab

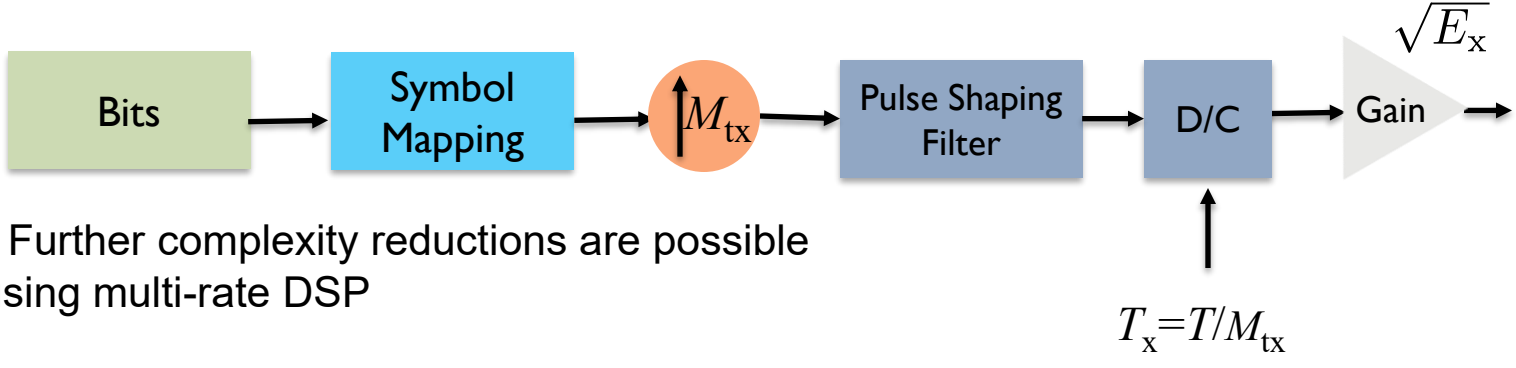
From Lecture #8

Transmitter structure

Theoretical block diagram



Practical block diagram

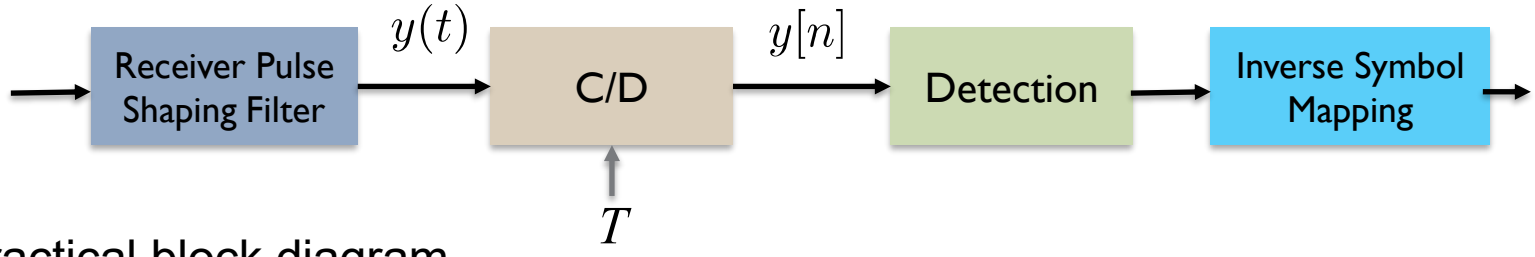


* Further complexity reductions are possible using multi-rate DSP

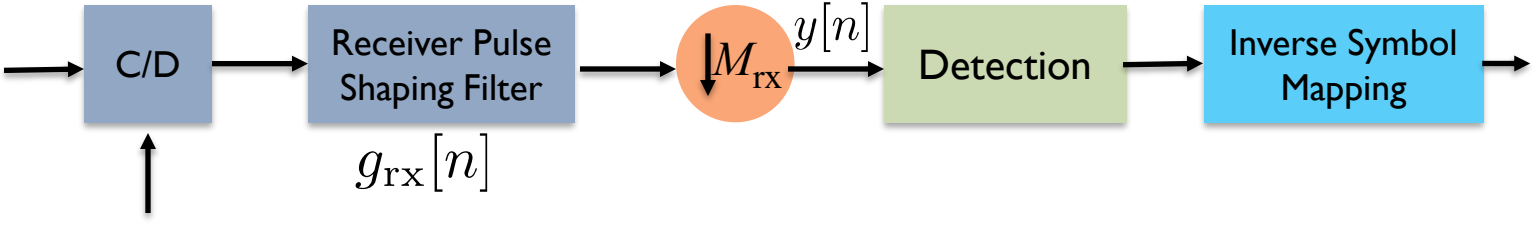
Preview of today´s lecture-I

Receiver structure

Theoretical block diagram

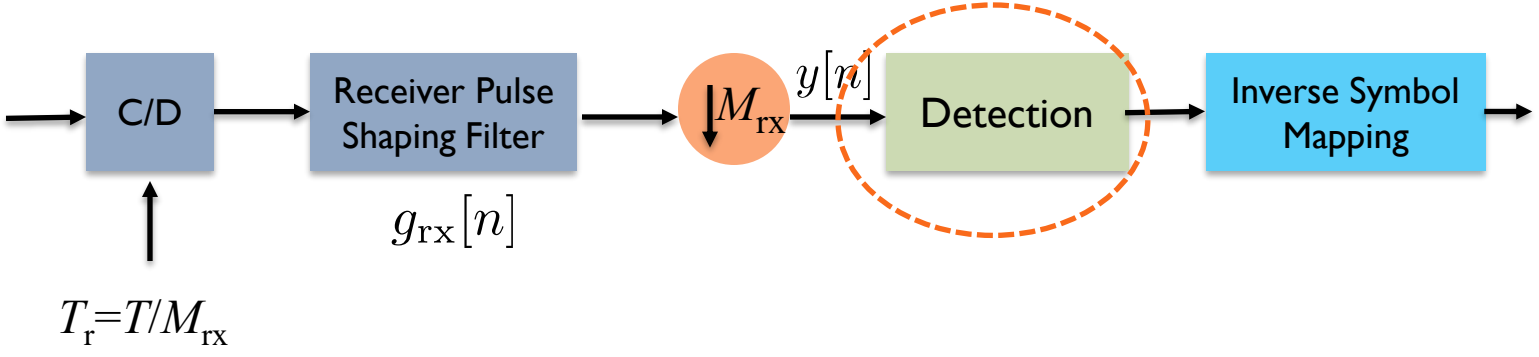


Practical block diagram



* Further complexity reductions are possible using multi-rate DSP

Preview of Today's Lecture-II



- Maximum likelihood (ML) detection
  - Formulate ML detection rule as an optimization problem
  - Define and plot Voronoi regions
- Probability of symbol error analysis
  - Derive the union bound on the symbol error probability
  - Compute the exact symbol error probability in special cases

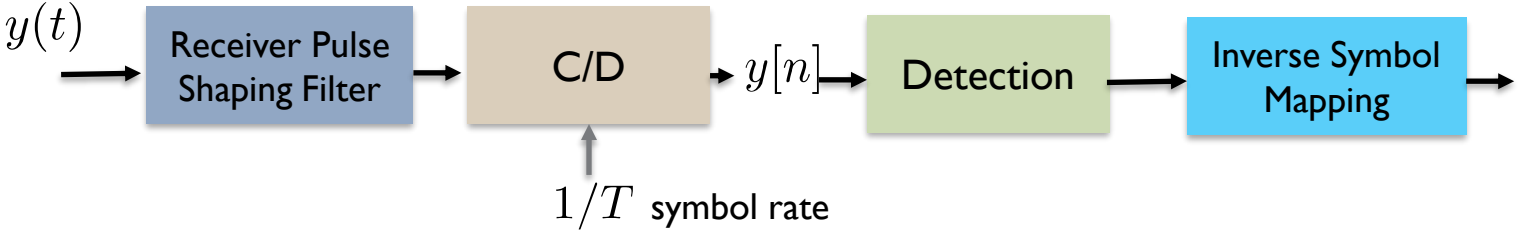
Digital Implementation of Receiving Matched Filter

Learning objectives:

- Implement discrete-time pulse shaping at the receiver using oversampling combined with downsampling
- Apply downsampling identities to implement and simplify receive pulse shaping

System model for the receiver

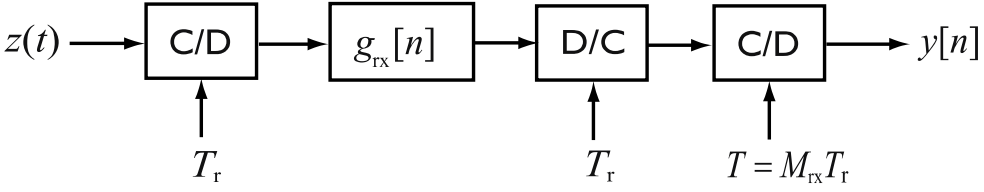
- Consider the following model for the receiver



- This structure does not map to a flexible implementation
  - Requires analog matched filtering
  - Does not take advantage of flexible digital processing
  - Does not lend itself to various synchronization tasks
- How can we implement pulse-shaping in discrete time?

Digital implementation of matched filter

- Discrete-time processing of a band limited continuous-time signal
  - Implement the analog filter using discrete-time processing



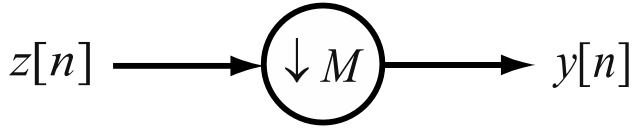
  - Let $T_r = T/M_{rx}$ for some integer M_{rx} then

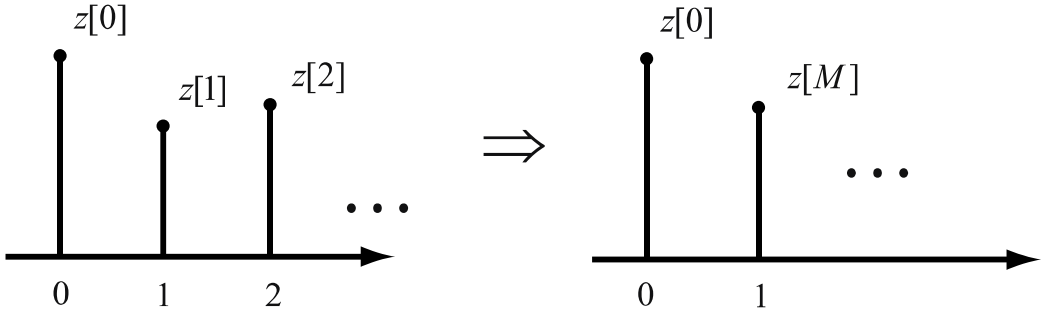
$$g_{rx}[n] = T_r g_{rx}(nT_r)$$

If $1/T_r$ > Nyquist rate, then this is called oversampling

  - Simplify using downsampling by M_{rx}

Downsampling





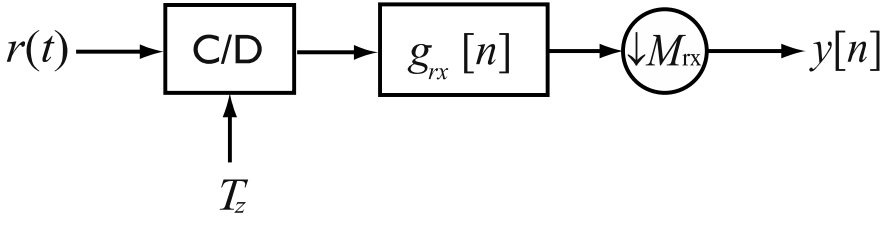
$$y[n] = z[nM]$$

$$Y(e^{j2\pi f}) = \frac{1}{M}\sum_{m=0}^{M-1} Z\left(e^{j2\pi f/M - j2\pi m/M}\right)$$

Digital implementation of matched filter

- If $r[n] = r(nT_{\mathrm{r}})$ we can simplify with downsampling

$$y[n] = \sum_{k=-\infty}^{\infty} g_{\mathrm{rx}}[k] r[nM_{\mathrm{rx}} - k]$$



- Note that lots of filtering computations are still wasted because we discard those coefficients when downsampling

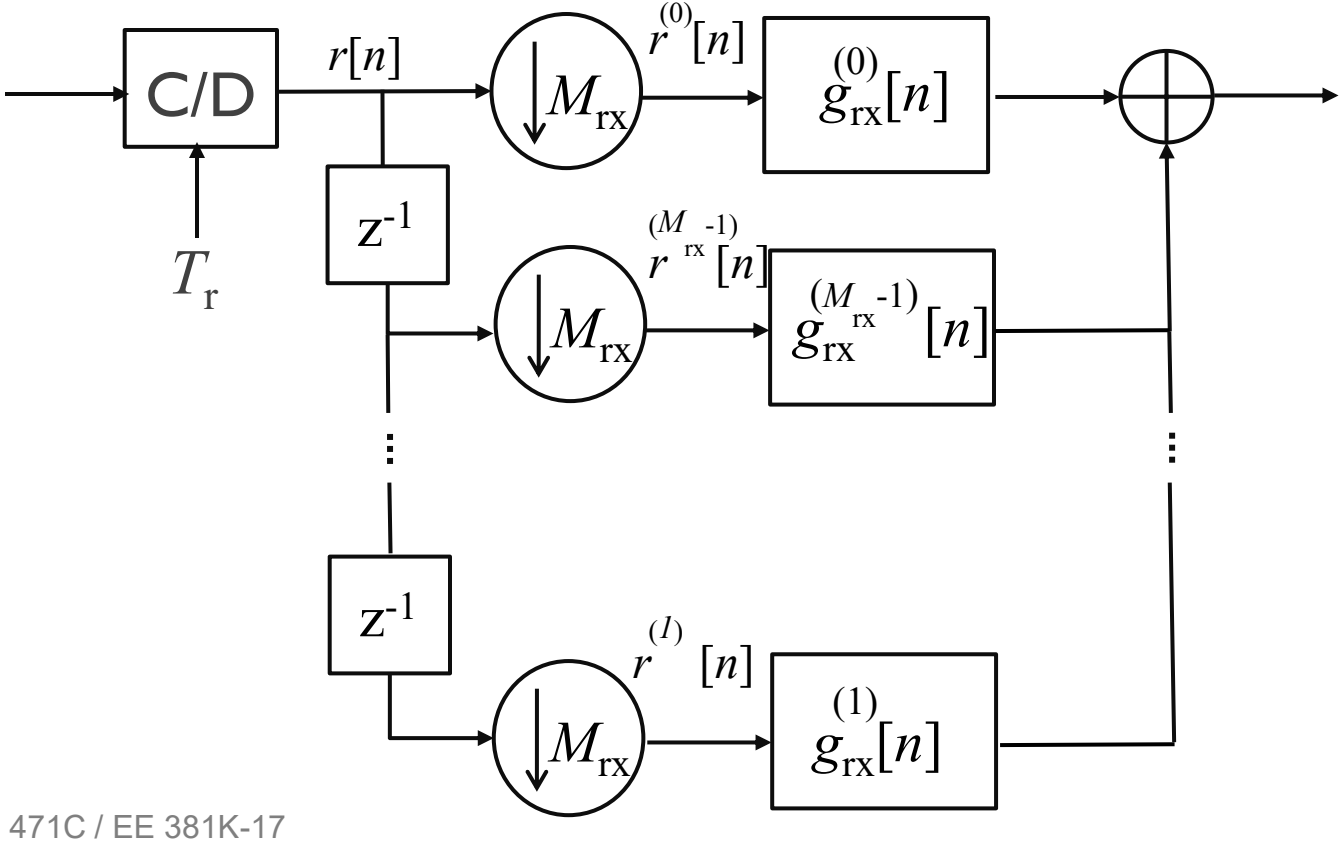
Digital implementation of matched filter

- Use downsampling equivalence to reduce computations

$$
\begin{aligned}
y[n] &= \sum_{k=-\infty}^{\infty} r[k] g_{\mathrm{rx}}[nM_{\mathrm{rx}} - k] \\
&= \sum_{p=-\infty}^{\infty} \sum_{m=0}^{M-1} r[pM_{\mathrm{rx}} + m] g_{\mathrm{rx}}[nM_{\mathrm{rx}} - pM_{\mathrm{rx}} - m] \\
&= \sum_{m=0}^{M_{\mathrm{rx}}-1} \sum_{p=-\infty}^{\infty} r^{(m)}[p] g_{\mathrm{rx}}[M_{\mathrm{rx}}(n-p) - m] \\
&= \sum_{m=0}^{M_{\mathrm{rx}}-1} \sum_{p=-\infty}^{\infty} r^{(m)}[p] g_{\mathrm{rx}}[M_{\mathrm{rx}}(n-p-1) + M_{\mathrm{rx}} - m] \\
&= \sum_{m=0}^{M_{\mathrm{rx}}-1} \sum_{p=-\infty}^{\infty} r^{(m)}[p] g_{\mathrm{rx}}^{(M_{\mathrm{rx}}-m)}[n-p-1]
\end{aligned}
$$

Digital implementation of matched filter

- We can finally exchange of downsampling and filtering operations



Digital Implementation of Receiving Matched Filter

Learning objectives:

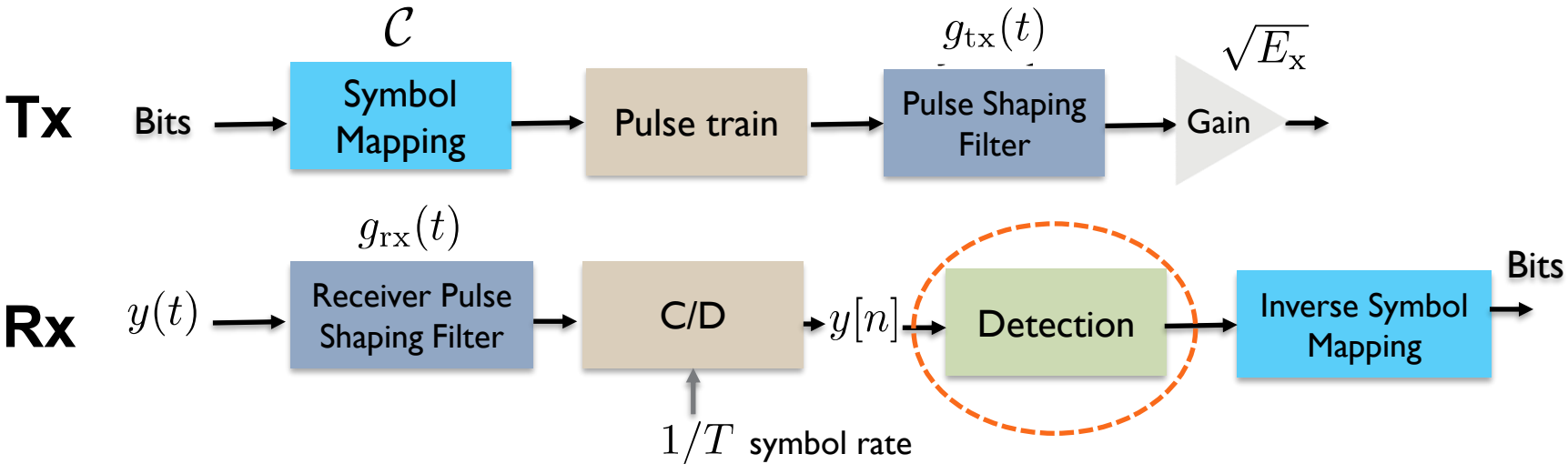
- Implement discrete-time pulse shaping at the receiver using oversampling combined with downsampling
- Apply downsampling identities to implement and simplify receive pulse shaping

Maximum likelihood detection

Learning objectives:

- Derive the ML detector for the AWGN channel
- Implement the ML detection algorithm

From previous lectures



- In this lecture we will focus on the optimal design of the detector

Symbol Detection in the AWGN Channel

- Using a matched filter and composite pulse shape satisfying Nyquist

$$y[n] = \sqrt{E_{\mathrm{x}}} \sum_{m} s[m] g((n-m)T) \; + \; g_{\mathrm{rx}}(t) \star v(t)|_{nT}$$

$$y[n] = \sqrt{E_{\mathrm{x}}} s[n] \; + \; v[n]$$

where $v[n]$ is IID complex Gaussian noise with $\mathcal{N}_c(0, N_0)$

Follows because $g_{rx}(t)$ is a matched filter and forms a Nyquist pulse-shape

Detection problem: find a good guess of $s[n]$ given an observation of $y[n]$

Maximum likelihood (ML) detection (1)

- This decision rule is optimal if symbols are equally likely
- Optimal detection $\Longleftrightarrow$ minimize probability of detection error
- ML maximizes the likelihood of the observation under the hypothesis that a given symbol was transmitted
- Likelihood function: $f_{y|s}(y[n]|s[n] = s)$

  - $f_{y|s}(x)$ is the conditional probability distribution function of $y[n]$ given $s[n]$

Maximum likelihood (ML) detection (2)

- The ML detector solves the optimization

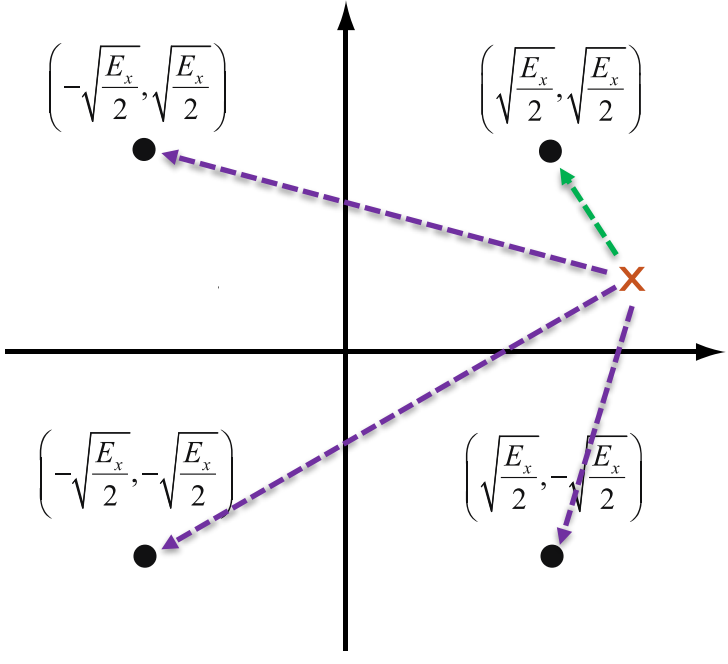
$$\begin{aligned}\arg\max_{s\in\mathcal{C}} f_{y|s}(y[n]|s[n]=s) &= \arg\max_{s\in\mathcal{C}} \frac{1}{\pi N_o} e^{-\frac{|y[n]-\sqrt{E_x}s|^2}{N_o}} \\ &= \arg\min_{s\in\mathcal{C}} e^{-\frac{|y[n]-\sqrt{E_x}s|^2}{N_o}} \\ &= \arg\min_{s\in\mathcal{C}} \frac{|y[n]-\sqrt{E_x}s|^2}{N_o} \\ &= \arg\min_{s\in\mathcal{C}} |y[n]-\sqrt{E_x}s|^2\end{aligned}$$

- ML detection for the AWGN channel thus solves

$$\arg\min_{s\in\mathcal{C}} |y[n]-\sqrt{E_x}s|^2$$

Maximum likelihood (ML) detection (3)

- ML detection is equivalent to choosing the closest symbol to the observation in terms of the squared error (Euclidean distance)



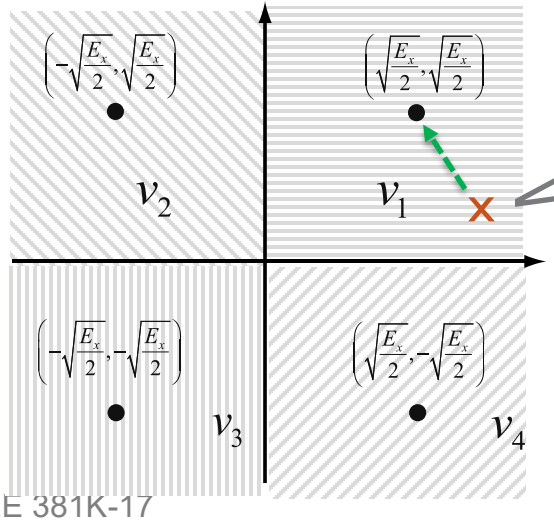
Example for 4-QAM

Voronoi regions

- Voronoi regions of the constellation $\mathcal{C}$ with AWGN detection

$$\mathcal{V}_{s_l} = \left\{ y : \left| y - \sqrt{E_x} s_l \right|^2 < \left| y - \sqrt{E_x} s_k \right|^2, \quad s_l, s_k \in C, \quad k \neq l \right\}$$

- Voronoi regions for the 4-QAM constellation are the quadrants



Any observation that falls in a Voronoi region is mapped to the corresponding symbol that generates that region through the ML detection process

Can use Voronoi regions to simplify the detection process

Maximum likelihood detection

Learning objectives:

- Derive the ML detector for the AWGN channel
- Implement the ML detection algorithm

Probability of symbol error analysis

Learning objectives:

- Derive the union bound on the probability of symbol error for the ML detector in an AWGN channel
- Calculate the probability of symbol error for QAM using both the union bound and the exact formula

Probability of symbol error

- Performance of a detector is measured by the probability of error
  - Can be computed in closed form in some cases
  - Estimated via bounds in most cases
- Probability of symbol error is relevant using symbol detection
  - Can also derive probability of bit error, frame error, etc
  - Probability of bit error is most widely used in commercial systems
- For equally likely symbols in AWGN the probability of symbol error

$$P_e\left(\frac{E_x}{N_o}\right) = \frac{1}{M}\sum_{m=0}^{M-1} P_{e|s_m}\left(\frac{E_x}{N_o}\right)$$

Computation of the probability of symbol error (1)

- It is a function of the SNR in Gaussian channels $\frac{E_x}{N_o}$

$$P_{e|s_m}\left(\frac{E_x}{N_o}\right) = \Pr\{s_m \text{is detected incorrectly} | s_m \text{was transmitted}\}$$

$$= \sum_{\substack{l=0 \\ l \neq m}}^{M-1} \Pr\{s_m \text{is decoded as } s_l | s_m \text{was transmitted}, m \neq l\}$$

- For complex AWGN

Gaussian Q-function

$$\mathbb{P}[s_m \rightarrow s_\ell] = Q\left(\sqrt{\frac{E_x}{N_o}\frac{|s_m - s_\ell|^2}{2}}\right)$$

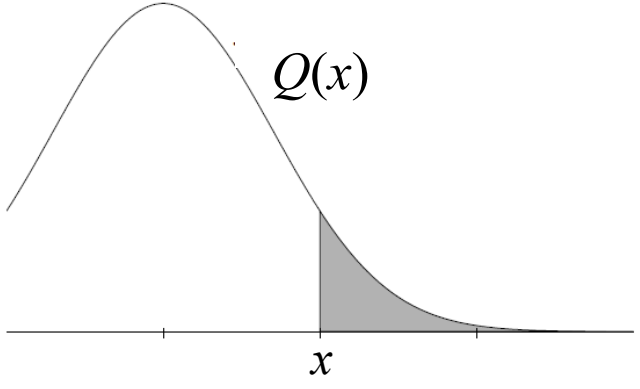
probability that s_m is decoded as s_l assuming that the constellation consists of two symbols

Computation of the probability of symbol error (2)

- Gaussian Q function

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$$

$$\leq \frac{1}{2} e^{-x^2/4}$$



Use the Chernoff bound to simplify or devise intuition in some cases

Union bound

- Probability of symbol error is hard to compute in practice
- We use a bound based on the pairwise error probability (PEP), defined

$$\mathbb{P}\left[s_m \to s_\ell\right] = \mathbb{P}\left[\left|y - \sqrt{E_{\mathrm{x}}} s_m\right|^2 > \left|y - \sqrt{E_{\mathrm{x}}} s_\ell\right|^2\right]$$

$$\mathbb{P}\left[s_m \to s_\ell\right] = Q\left(\sqrt{\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}} \frac{\left|s_m - s_\ell\right|^2}{2}}\right)$$ ← For complex AWGN

- The PEP assumes only two symbols in the constellation, thus provides an upper bound since the region of possible error is larger

$$\mathbb{P}\left[s_m \text{is decoded as } s_\ell | s_m \text{was transmitted}, m \neq l\right] \leqslant \mathbb{P}\left[s_m \to s_\ell\right]$$

Computing P_e using the union bound

- Using the union bound on the probability of error

$$P_{\mathrm{e}|s_m}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}\right) \leqslant \sum_{\substack{\ell=0 \\ \ell \neq m}}^{M-1} \mathbb{P}\left[s_m \to s_\ell\right]$$

- Substituting into the bound

$$P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}\right) \leqslant \frac{1}{M} \sum_{m=0}^{M-1} \sum_{\ell=0, \ell \neq m}^{M-1} Q\left(\sqrt{\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}} \frac{\left|s_m - s_\ell\right|^2}{2}}\right)$$

Computing P_e in terms of d_{min}

- Let $d^2_{\min} = \min_{s_\ell \in \mathcal{C}, s_m \in \mathcal{C}, s_m \neq s_\ell} |s_\ell - s_m|^2$
- The minimum distance provides a lower bound for any distinct pair of symbols s_l and s_m

$$d^2_{\min} \leqslant |s_\ell - s_m|^2$$

- Using the minimum distance

$$P_e\left(\frac{E_x}{N_o}\right) \leq \frac{1}{M}\sum_{m=0}^{M-1}(M-1)Q\left(\sqrt{\frac{E_x}{N_o}\frac{d^2_{\min}}{2}}\right)$$

$$= (M-1)Q\left(\sqrt{\frac{E_x}{N_o}\frac{d^2_{\min}}{2}}\right)$$

UNION BOUND
(loose at low SNR)

Union bound for M-QAM

- For M-QAM

$$d_{\min}^2 = \frac{6}{M-1}$$

- Substituting into the general equation for the union bound

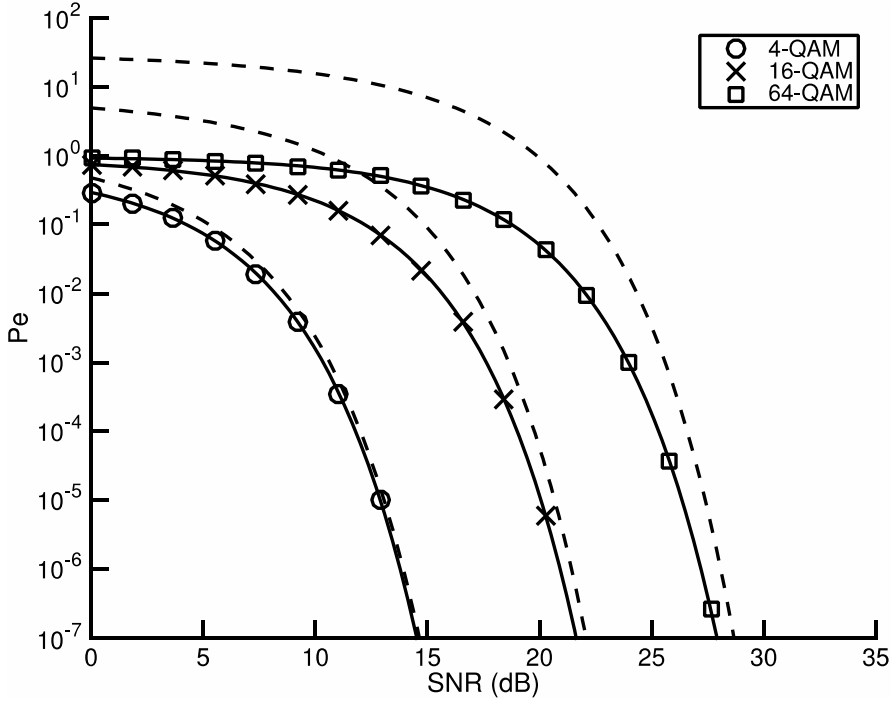
$$P_e^{\text{QAM}}\left(\frac{E_x}{N_o}\right) \leq (M-1)Q\left(\sqrt{\frac{E_x}{N_o}\frac{3}{M-1}}\right)$$

- Exact calculation is possible (detailed derivation not required)

$$P_e^{\text{QAM}}\left(\frac{E_x}{N_o}\right) = 4\left(1-\frac{1}{\sqrt{M}}\right)Q\left(\sqrt{\frac{E_x}{N_o}\frac{3}{M-1}}\right) - 4\left(1-\frac{1}{\sqrt{M}}\right)^2\left[Q\left(\sqrt{\frac{E_x}{N_o}\frac{3}{M-1}}\right)\right]^2$$

Example

- Union bound and exact probability of error for M-QAM



Observations

- The probability of error for an AWGN channel is a function of the SNR and shows a waterfall effect
- The probability of error is a useful tool in the design and analysis of communication networks
- The application dictates a target bit error probability after error control decoding
  - It can be translated into an effective probability of symbol error before decoding
- Make choices about the transmit power and RF components to achieve this design target

Probability of symbol error analysis

Learning objectives:

- Derive the union bound on the probability of symbol error for the ML detector in an AWGN channel
- Calculate the probability of symbol error for QAM using both the union bound and the exact formula